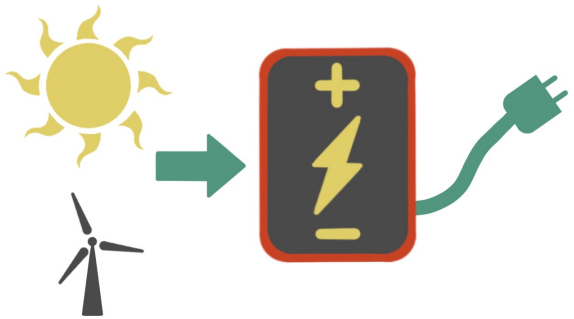
BMS Temperature Measurement



Ania's BMS Course

by Ania Mitros, PhD

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https://ofb.net/~ania/Anias-BMS-course/

Overview: Temperature Measurement

- Why measure temperature?
- Sensor placement
- Thermistor vs thermocouple
- Thermistor circuit & equations

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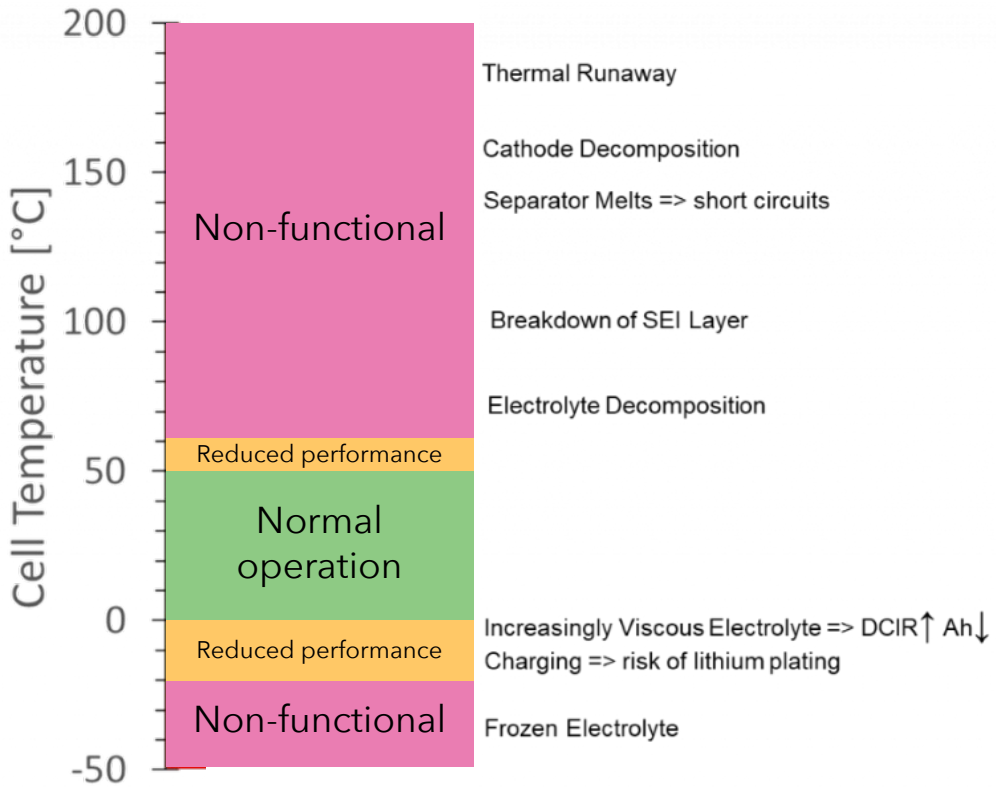
Why measure temperature?

Do measure temperature for:

1. **Controls:** Require decreased current at hot/cold
2. **Safety:** Detect over-temperature to issue user alert and/or initiate mitigation.

Don't measure temperature for:

1. **Small systems:** Inexpensive system and/or low consequence to over-temperature.



https://www.batterydesign.net/cell-temperature-sensing/

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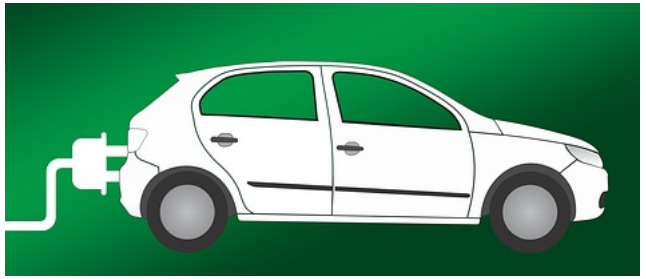
What systems measure temperature?

Unlike cell voltage sensing, battery temperature sensing may be omitted.

Electric car



✓ Controls: Charge rate, max power, cooling/heating system

✓ Safety: Large battery carries large consequences

✓ Cost: Temperature sensor cost is small fraction of system

Home energy

✓ Controls: Like car

✓ Safety: Large battery, possibly sleeping or immobile home occupants

✓ Cost: Like car

Small drone

✗ Controls: External charger

✗ Safety: Small battery charged outside system

✗ Cost: Cheap so temperature sensing would be substantial increase with additional impact to weight

https://pixabay.com/illustrations/electric-car-gas-station-environment-2728131/
https://pixabay.com/photos/drones-small-quadcopter-toy-remote-1620464/

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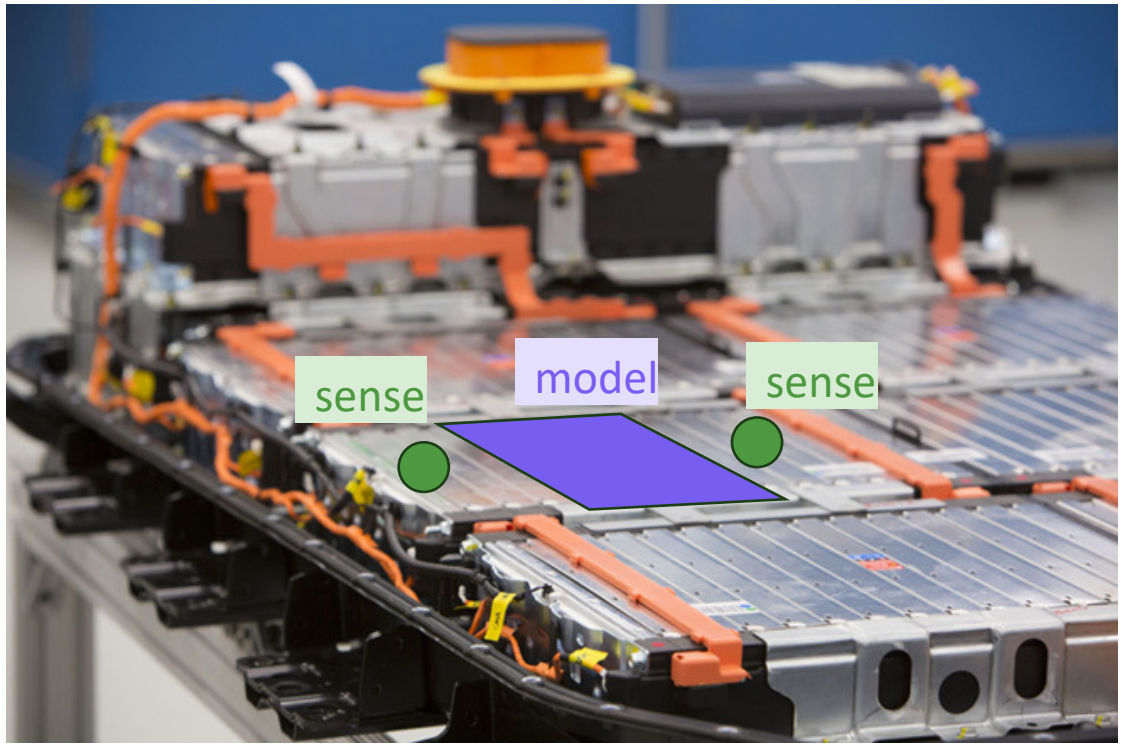
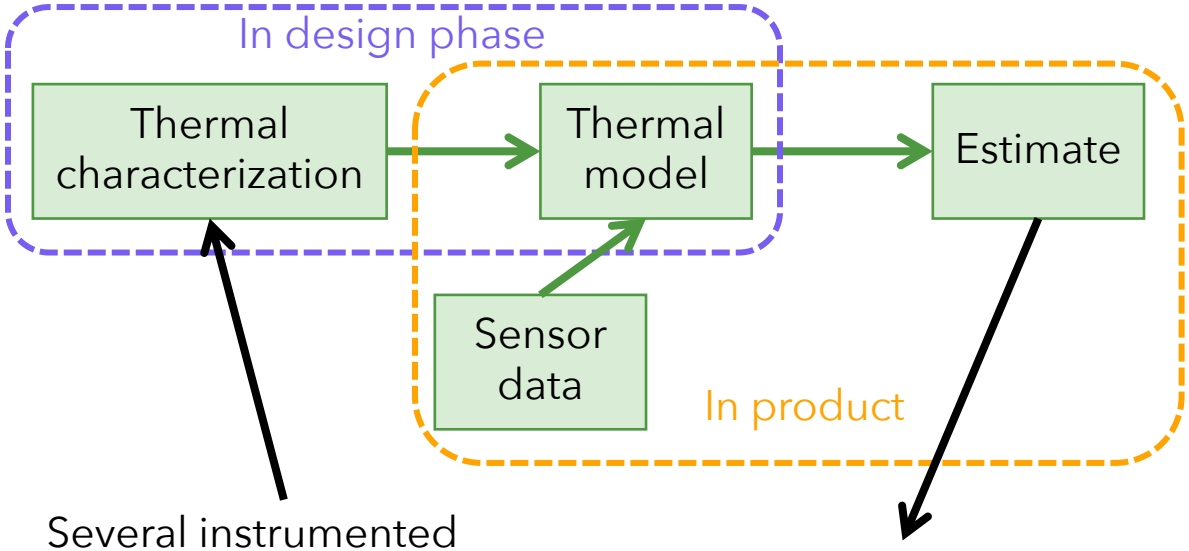
Temperature sensor placement

Do we need a sensor on every cell?



Controls:



Several instrumented batteries charged & discharged at several temperatures

Residual error from:

- Model imperfection
- Variation between units

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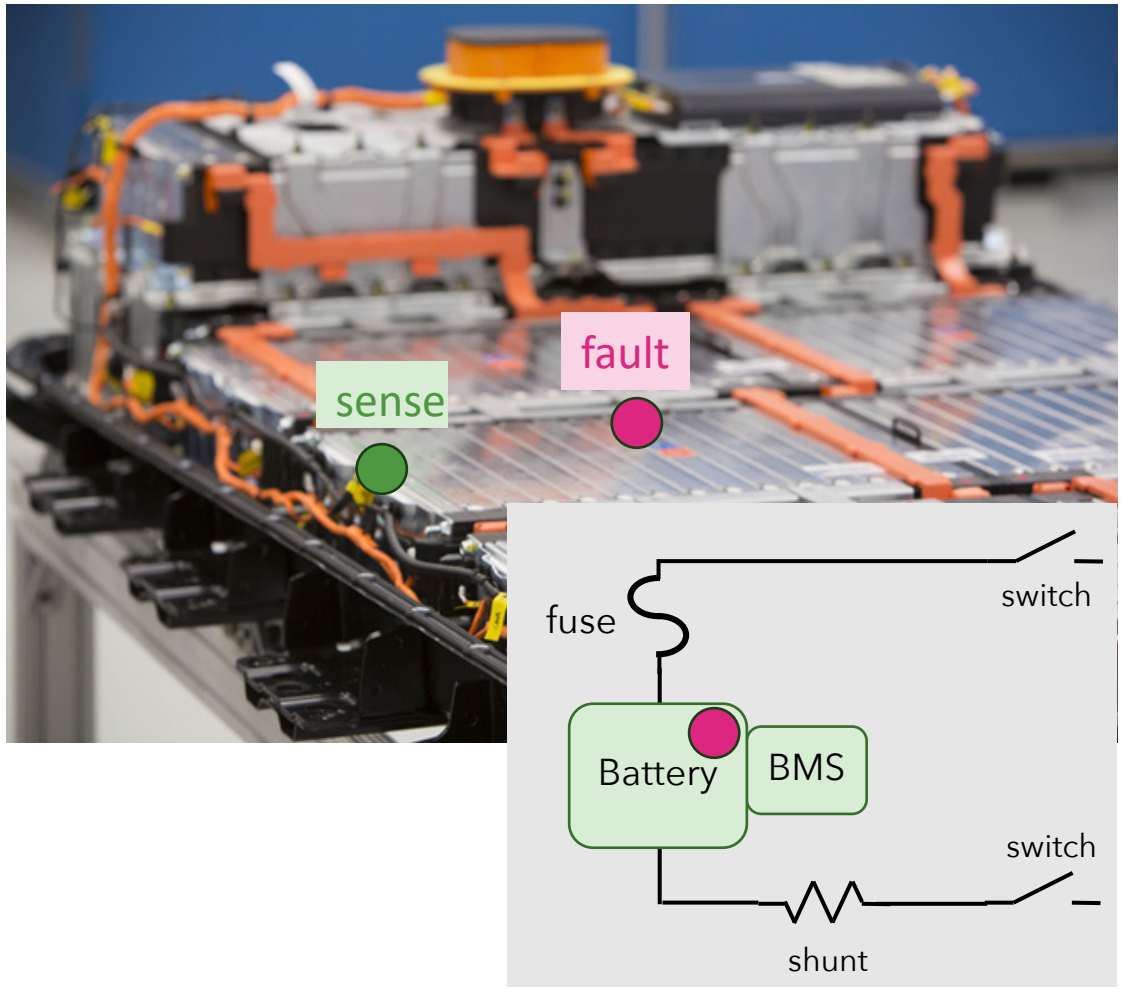
Temperature sensor placement

Do we need a sensor on every cell?



Safety:

Warm battery → Reduce current

Cold battery → Reduce current

← Controls scenario

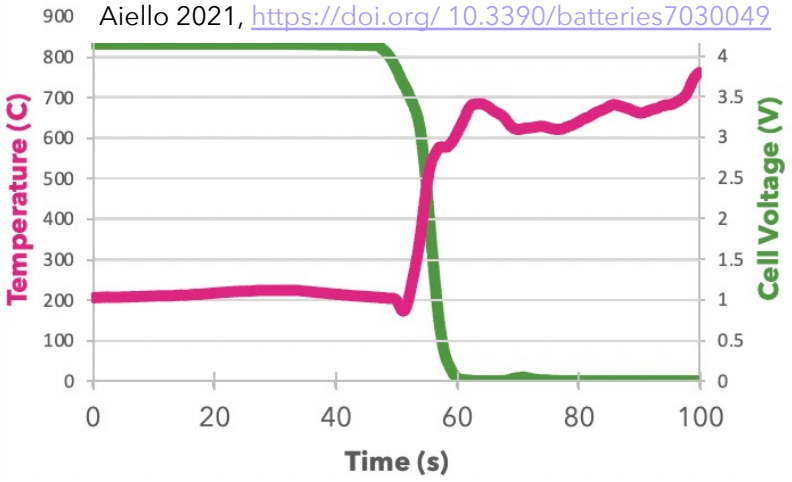
Exothermic behavior? Fire?

→ Need sensor on every cell to detect quickly

→ BMS can't do much. It's too late.

→ Can use cell voltage

Aiello 2021, https://doi.org/ 10.3390/batteries7030049



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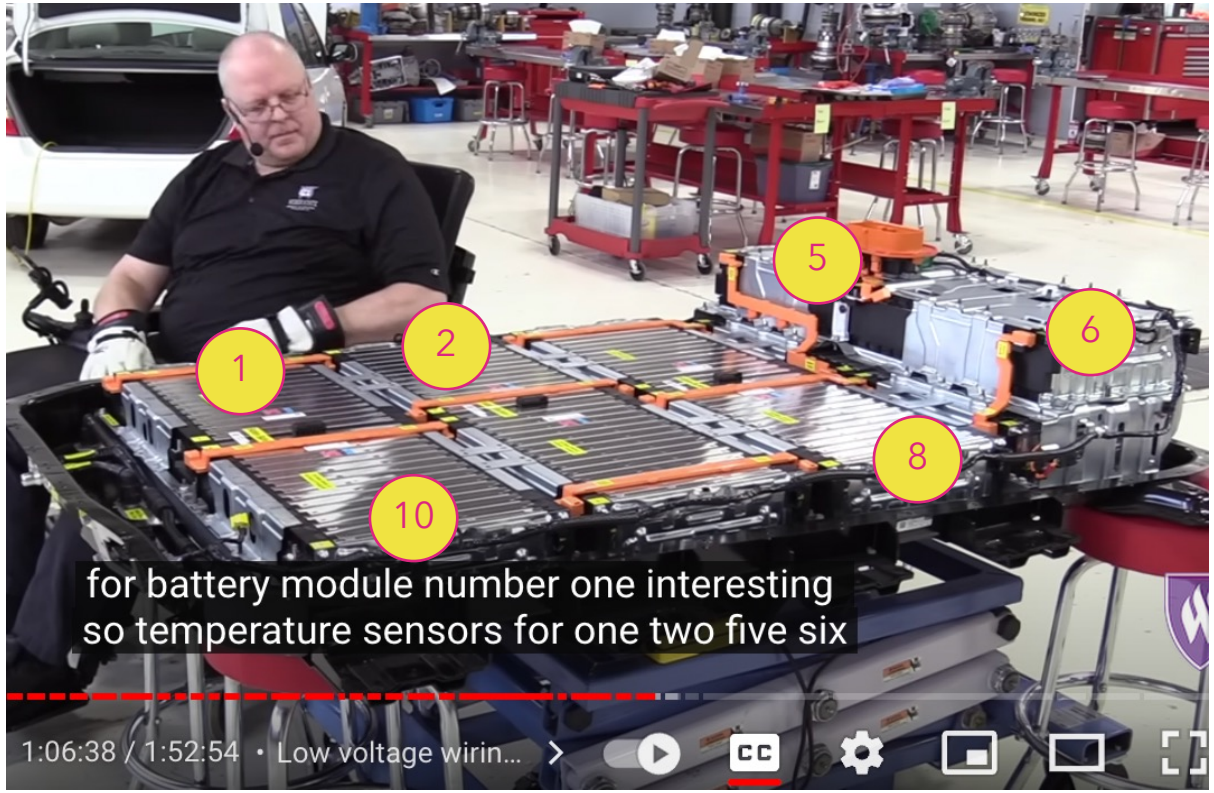
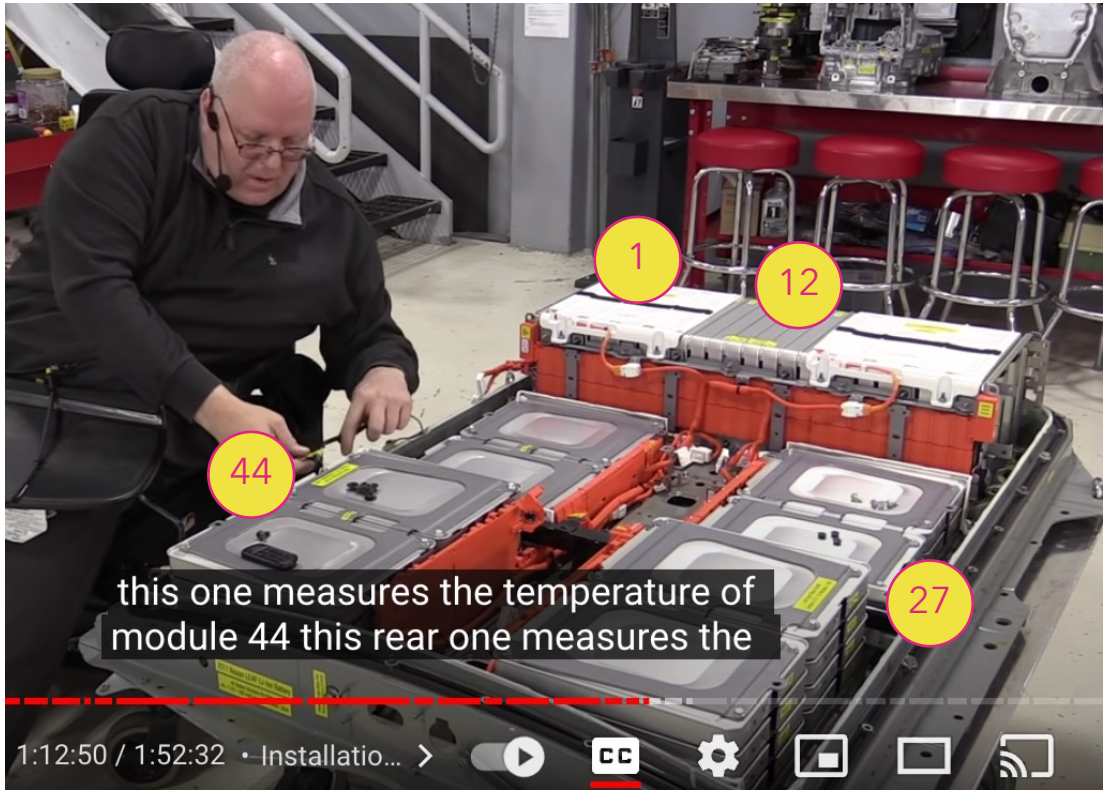
Temperature sensor placement: Examples



Chevy Bolt
Thermal sensors on ends of modules 1, 2, 5, 6, 8 & 10

WeberAuto
https://www.youtube.com/watch?v=ZBzRKqlr95U&t=3998s



2011 Nissan LEAF Battery
Thermal sensors on modules 44, 1, 12, 27

WeberAuto
https://www.youtube.com/watch?v=vYQJatWpBXY

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Sensor placement summary

~~**Sensor on every cell**~~

- Safety: Can detect thermal runaway

… but so can individual cell voltage sensing

A few sensors for whole battery pack

- Controls: Can serve as input to controls algorithms which model temperatures between sensors to determine cooling needs, heating needs, and max allowable current.
- Typical solution

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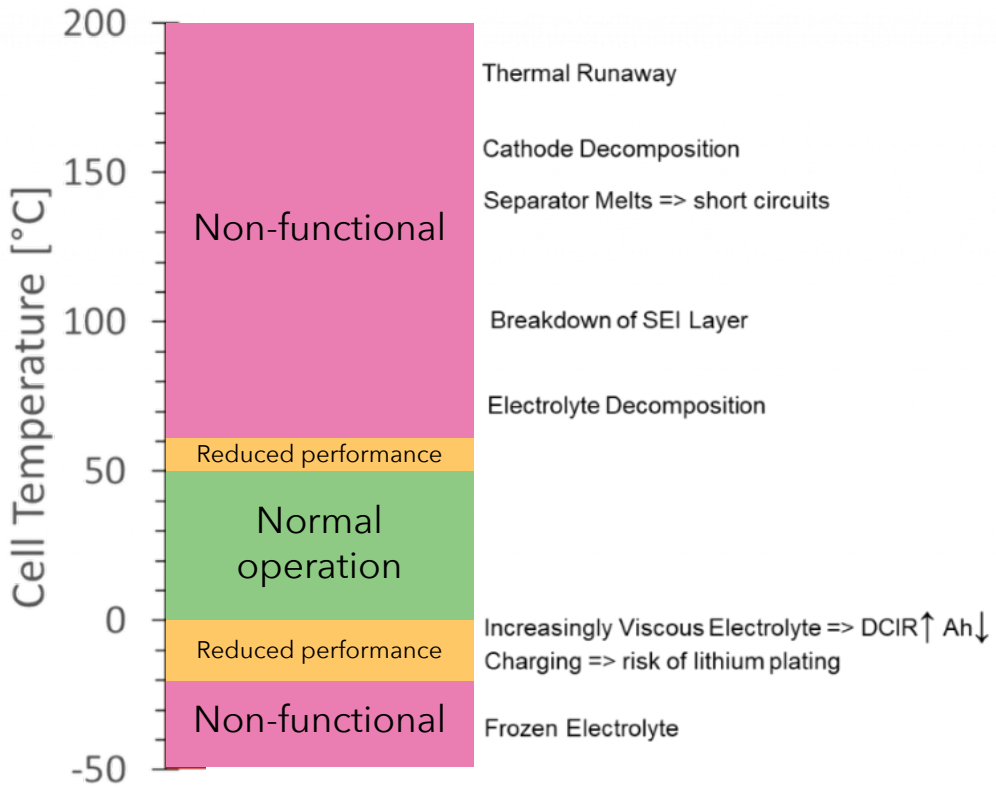

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Thermistor vs thermocouple

	Thermistor	Thermocouple
Construction	Thermally sensitive resistor	Variable voltage created by two dissimilar metals
Temperature range	-50 to 250 °C ✓	-200 to 1750 °C
Drift	0.2 °C per year (or better with hermetical sealing) ✓	1-2 °C per year
Accuracy	± 0.05°C at best	±1°C at best
Response time	Similar	
Cost	Pennies ✓ ✓ ✓	Dollars

Learn more: https://www.ametherm.com/blog/thermistors/temperature-sensors-thermistors-vs-thermocouples

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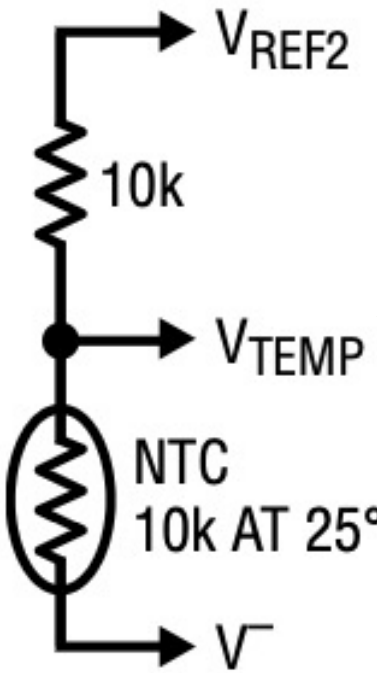
Thermistor circuit

Stable reference voltage

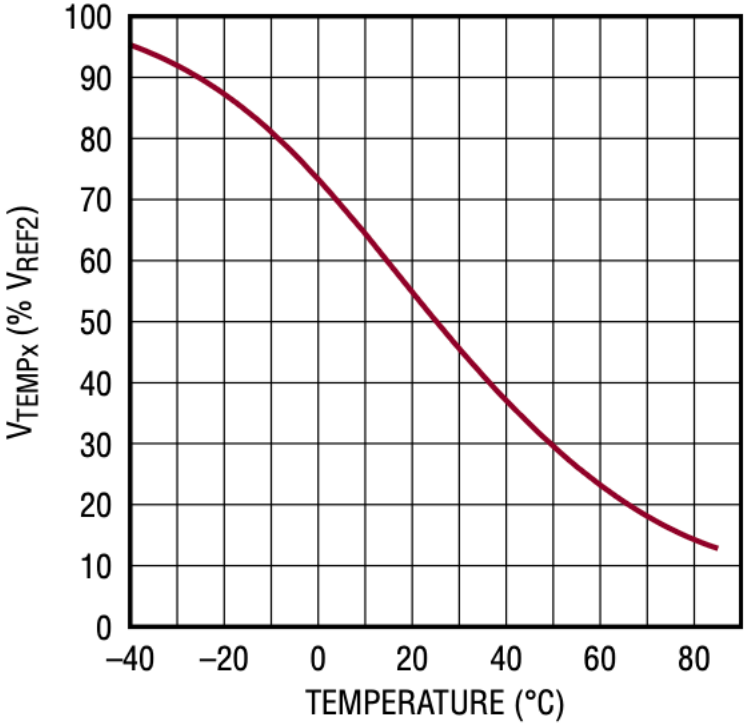


Sensor measurement

NTC = Negative Temperature Coefficient thermistor

Ground

$$V_{TEMP} = V_{REF2} \times \frac{R_{NTC}}{R_{NTC} + R_{10k}}$$



As temperature ↑

R_{NTC} ↓

$$\downarrow V_{TEMP} = V_{REF2} \times \frac{\downarrow R_{NTC}}{\downarrow R_{NTC} + R_{10k}}$$

Circuit and curve from LTC6812-1 datasheet

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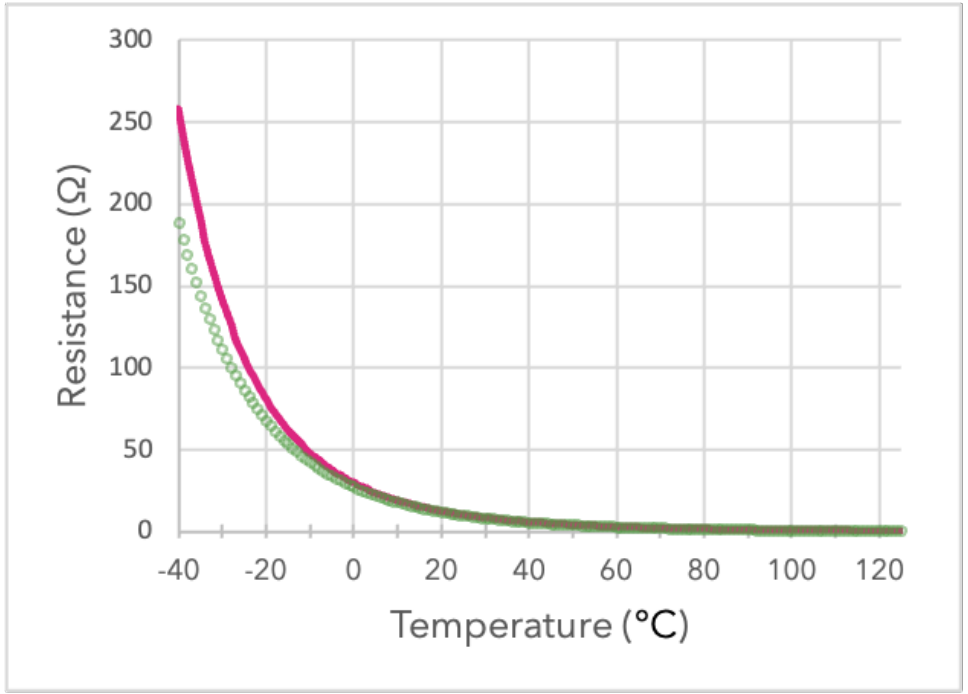
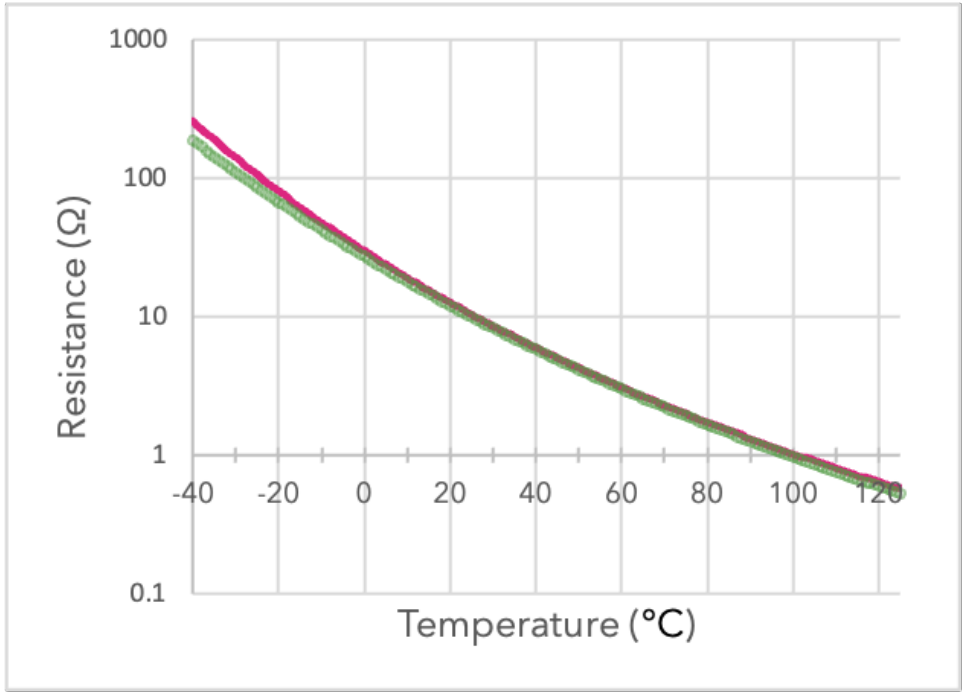
Thermistor: R vs temperature





NTC thermistor from TDK
Surface mount 0603
R_0 = 10kΩ at 25°C
B value [25/85°C] = 3435K
±1% tolerance, and ±1% beta tolerance

$$R(T) = R_0 \cdot e^{\left(\beta \cdot \left(\frac{1}{T} - \frac{1}{T_0}\right)\right)}$$

R_0 =10kΩ

β =3435K

T_0 =25°C=298.15K

Data from: https://product.tdk.com/en/search/sensor/ntc/chip-ntc-thermistor/info?part_no=NTCG063JF103FTB

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Practical implementation

$$R_{NTC} = R(T) = R_0 \cdot e^{\left(\beta \cdot \left(\frac{1}{T} - \frac{1}{T_0}\right)\right)}$$

$$V_{TEMP} = V_{REF2} \times \frac{R_{NTC}}{R_{NTC} + R_{10k}}$$

Tolerances (from datasheets):

- R_0 and Beta
- Voltage measurement

T(°C)	R_{NTC}	R_{NTC}+1%	V_{TEMP}	V_{TEMP} +V_{ERR}	…	T(°C) meas'd
-40						
-30						
-20						
-10						
0						
20						
…						

Requirement met?

Thermal model → Requirements: (example)
±1°C for 0°C to 50°C
±2°C for -40°C to 125°C

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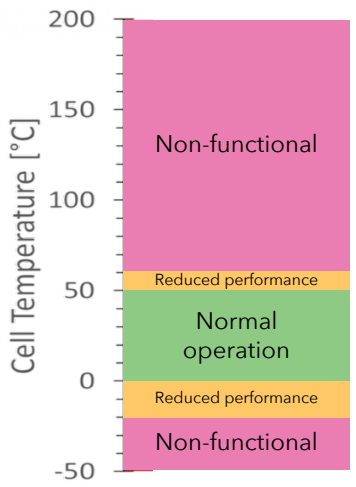
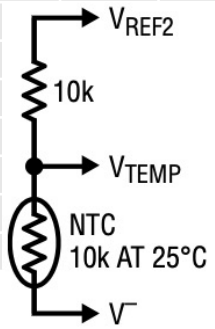
Multidisciplinary design



T(°C)	RNTC	RNTC+1 %	VTEMP	VTEMP +VERR	...	T(°C) meas'd
-40						
-30						
-20						
-10						
0						
10						
20						
...						



Thermal characterization

Thermal model

Thermal/Mechanical Engineers

Systems/Controls Engineers

Cell Engineers

Electrical Engineers

Optimized design

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"The End" by Gerd Altmann from Pixabay